

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security

Practical 1 A

Aim : Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques cli

Code :

```
c=lambda t,s: "".join(chr((ord(x)-(65 if x.isupper() else 97)+s)%26+(65 if x.isupper() else 97)) if
x.isalpha() else x for x in t)
```

```
m=lambda t,k,d=0: "".join(((k[i]:chr(i+97) for i in range(26)) if d else k)[x.lower()].upper() if x.isupper()
else ((k[i]:chr(i+97) for i in range(26)) if d else k)[x.lower()]) if x.isalpha() else x for x in t)
```

```
t=input("Msg: ")
ch=input("1.C 2.M: ")
```

```
if ch=="1":
    s=int(input("Shift: "))
    e=c(t,s)
    print("Enc:", e)
    print("Dec:", c(e,-s))
```

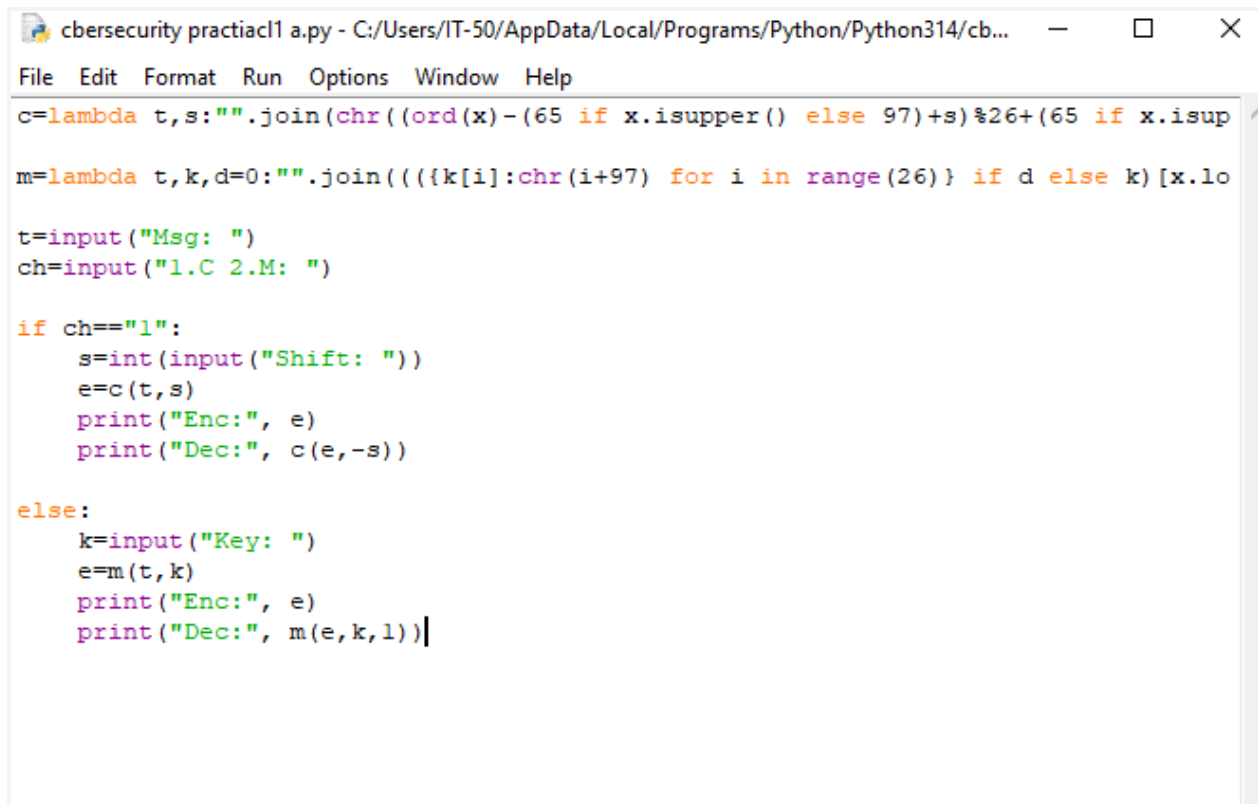
```
else:
    k=input("Key: ")
    e=m(t,k)
    print("Enc:", e)
    print("Dec:", m(e,k,1))
```

SCREEN SHOTS :

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Practical 1 A



```
cbersecurity practiac1 a.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cb...
File Edit Format Run Options Window Help
c=lambda t,s:"".join(chr((ord(x)-(65 if x.isupper() else 97)+s)%26+(65 if x.isup
m=lambda t,k,d=0:"".join(((k[i]:chr(i+97) for i in range(26)} if d else k)[x.lo

t=input("Msg: ")
ch=input("1.C 2.M: ")

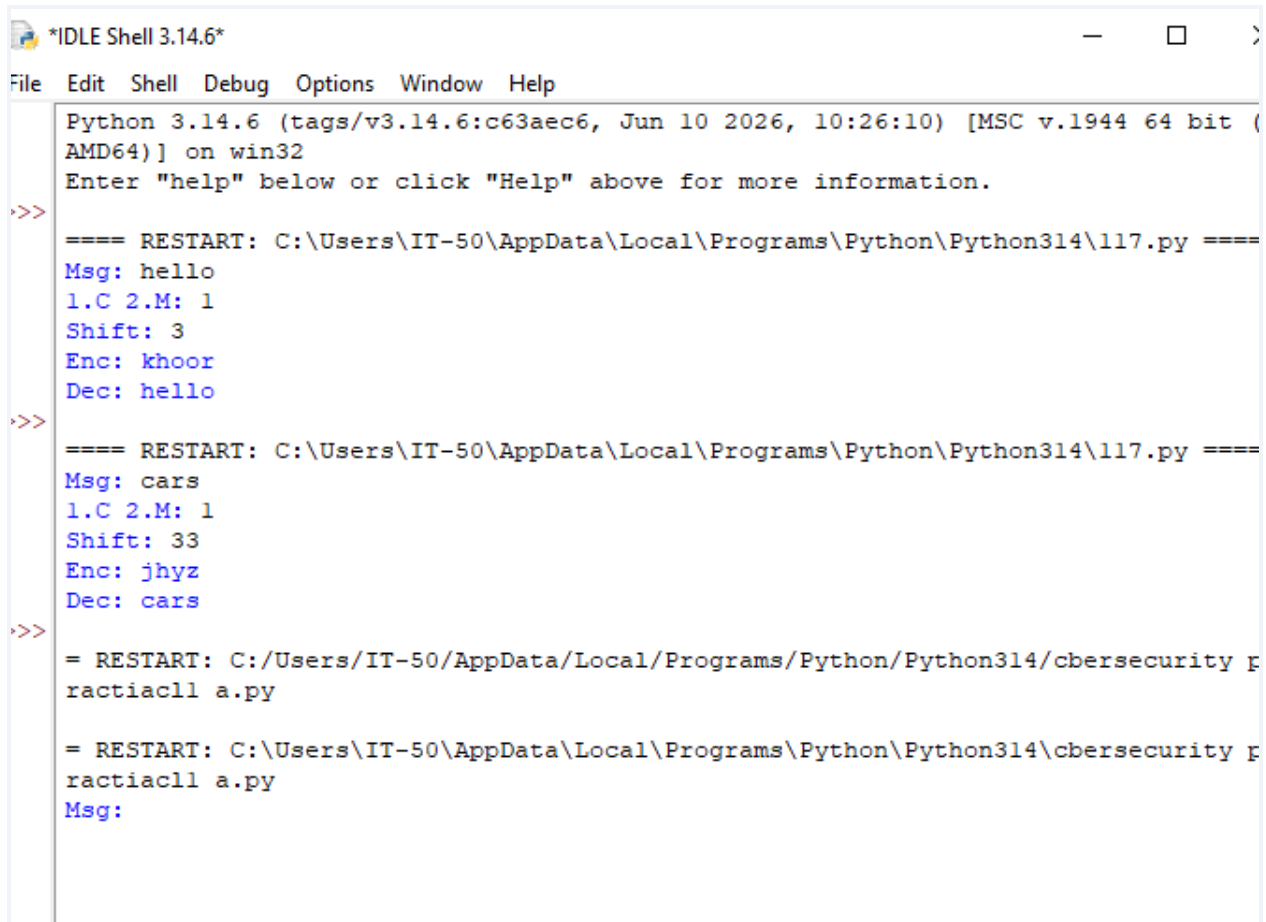
if ch=="1":
    s=int(input("Shift: "))
    e=c(t,s)
    print("Enc:", e)
    print("Dec:", c(e,-s))

else:
    k=input("Key: ")
    e=m(t,k)
    print("Enc:", e)
    print("Dec:", m(e,k,1))
```

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```
*IDLE Shell 3.14.6*
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:\Users\IT-50\AppData\Local\Programs\Python\Python314\117.py ====
Msg: hello
1.C 2.M: 1
Shift: 3
Enc: khoor
Dec: hello
>>>
==== RESTART: C:\Users\IT-50\AppData\Local\Programs\Python\Python314\117.py ====
Msg: cars
1.C 2.M: 1
Shift: 33
Enc: jhyz
Dec: cars
>>>
= RESTART: C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cbersecurity p
ractiac11 a.py
= RESTART: C:\Users\IT-50\AppData\Local\Programs\Python\Python314\cbersecurity p
ractiac11 a.py
Msg:
```

Output :

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\IT-50>"C:\Users\IT-50\AppData\Local\Programs\Python\Python314\cbersecurity practiac11 a.py"
Msg: hello
1.C 2.M: 1
Shift: 4
Enc: lipps
Dec: hello
```

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Practical 1 A

Aim : Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques gui

Code :

```
import tkinter as tk
```

```
c=lambda t,s: "".join(chr((ord(x)-(65 if x.isupper() else 97)+s)%26+(65 if x.isupper() else 97)) if
x.isalpha() else x for x in t)
m=lambda t,k,d=0: "".join(((k[i]:chr(i+97) for i in range(26)) if d else k)[x.lower()].upper() if x.isupper()
else ((k[i]:chr(i+97) for i in range(26)) if d else k)[x.lower()]) if x.isalpha() else x for x in t)
```

```
def run():
    t=text.get()
    if mode.get()=="Caesar":
        s=int(key.get())
        e=c(t,s)
        out.set(f"Enc: {e}\nDec: {c(e,-s)}")
    else:
        k=key.get()
        e=m(t,k)
        out.set(f"Enc: {e}\nDec: {m(e,k,1)}")
```

```
root=tk.Tk()
root.title("Cipher")
```

```
mode=tk.StringVar(value="Caesar")
key=tk.StringVar()
text=tk.StringVar()
out=tk.StringVar()
```

```
tk.Entry(root,textvariable=mode).pack()
tk.Entry(root,textvariable=key).pack()
tk.Entry(root,textvariable=text).pack()
tk.Button(root,text="Run",command=run).pack()
tk.Label(root,textvariable=out).pack()
```

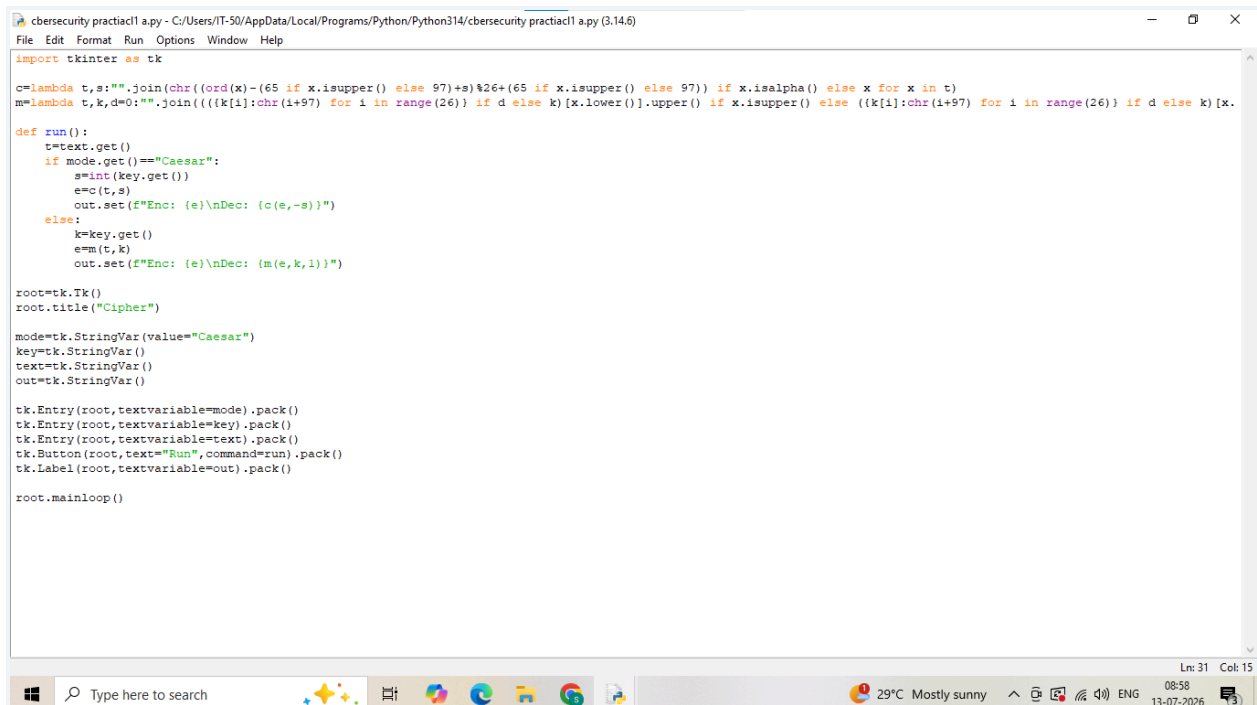
```
root.mainloop()
```

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SCREEN SHOT



```
import tkinter as tk

c=lambda t,s:"".join(chr((ord(x)-(65 if x.isupper() else 97)+s)%26+(65 if x.isupper() else 97)) if x.isalpha() else x for x in t)
m=lambda t,k,d="": "".join((((k[i]:chr(i+97) for i in range(26)) if d else k)[x.lower()].upper() if x.isupper() else ((k[i]:chr(i+97) for i in range(26)) if d else k)[x.

def run():
    t=text.get()
    if mode.get()=="Caesar":
        s=int(key.get())
        e=c(t,s)
        out.set(f"Enc: {e}\nDec: {c(e,-s)}")
    else:
        k=key.get()
        e=m(t,k)
        out.set(f"Enc: {e}\nDec: {m(e,k,1)}")

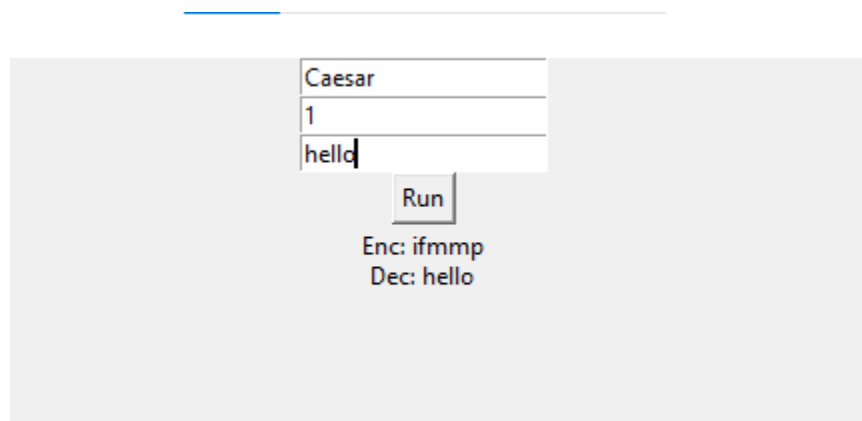
root=tk.Tk()
root.title("Cipher")

mode=tk.StringVar(value="Caesar")
key=tk.StringVar()
text=tk.StringVar()
out=tk.StringVar()

tk.Entry(root,textvariable=mode).pack()
tk.Entry(root,textvariable=key).pack()
tk.Entry(root,textvariable=text).pack()
tk.Button(root,text="Run",command=run).pack()
tk.Label(root,textvariable=out).pack()

root.mainloop()
```

Output



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